

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

Name	Damiya Aina binti Basir Abd Shammad
IC No. / Matric No.	040302030608 / A23CS0220
Year / Program	2 / SECPH
Section	02
Lecturer Name	Dr Sean Choon Sen
Case Study	ALK module
System Name	Koperasi KADA Membership & Loan Application System
Group Name	CO-T

SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

2 DDL SQL Statements

```
CREATE TABLE Member_Application (  
    applicant_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(255) NOT NULL,  
    id_number VARCHAR(20) NOT NULL UNIQUE,  
    email VARCHAR(255) NOT NULL,  
    marital_status ENUM('Bujang', 'Berkahwin', 'Duda', 'Janda') NOT  
NULL,  
    gender ENUM('Lelaki', 'Perempuan') NOT NULL,  
    religion VARCHAR(50) NOT NULL,  
    ethnicity VARCHAR(50) NOT NULL,  
    home_address TEXT NOT NULL,  
    postcode VARCHAR(10) NOT NULL,  
    state VARCHAR(50) NOT NULL,  
    membership_number VARCHAR(50) NOT NULL UNIQUE,  
    phone_mobile VARCHAR(20) NOT NULL,  
    monthly_salary DECIMAL(10, 2) NOT NULL  
);
```

```
CREATE TABLE Loan_Application (  
    loan_id INT AUTO_INCREMENT PRIMARY KEY,  
    member_id INT,  
    loan_type ENUM('Al-Bai', 'Al-Inah', 'Skim Khas', 'Karnival Musim  
Istimewa', 'Baik Pulih Kenderaan', 'Cukai Jalan', 'Lain-Lain'),  
    loan_amount DECIMAL(15, 2) NOT NULL,  
    repayment_period_months INT NOT NULL,  
    monthly_installment DECIMAL(15, 2) NOT NULL,  
    approval ENUM('Pending', 'Reviewed', 'Disapproved', 'Approved')  
DEFAULT 'Pending',
```

FOREIGN KEY (member_id) REFERENCES

Member_Application(applicant_id) ON DELETE CASCADE

);

8 DML SQL Statements

1. Insert a New Member into Member_Application

INSERT INTO Member_Application (

name, id_number, email, marital_status, gender, religion, ethnicity,

home_address,

postcode, state, membership_number, phone_mobile, monthly_salary

)

VALUES (

'Ahmad bin Ali', '760309030563', 'ahmad03@gmail.com', 'Berkahwin',

'Lelaki',

'Islam', 'Melayu', 'Lot 56, Kg Kubang Pasu', '17600', 'Kelantan', 'M023',

'0116783439', 8000.00

);

2. Insert a Loan Application into Loan_Application

INSERT INTO Loan_Application (

member_id, loan_type, loan_amount, repayment_period_months,

monthly_installment, approval

)

VALUES (1, 'Al-Bai', 10000.00, 24, 416.67, 'Pending'

);

3. Update the Loan Approval Status

UPDATE Loan_Application

SET approval = 'Approved'

WHERE loan_id = 1;

4. Delete a Member and Cascade their Loans

DELETE FROM Member_Application

WHERE applicant_id = 1;

5. Retrieve All Members with Active Loans (JOIN Query)

```
SELECT m.name, m.id_number, l.loan_type, l.loan_amount, l.approval
FROM Member_Application m
JOIN Loan_Application l ON m.applicant_id = l.member_id
WHERE l.approval = 'Pending';
```

6. Retrieve Total Loan Amounts for Each Member

```
SELECT m.name, SUM(l.loan_amount) AS total_loans
FROM Member_Application m
JOIN Loan_Application l ON m.applicant_id = l.member_id
GROUP BY m.name;
```

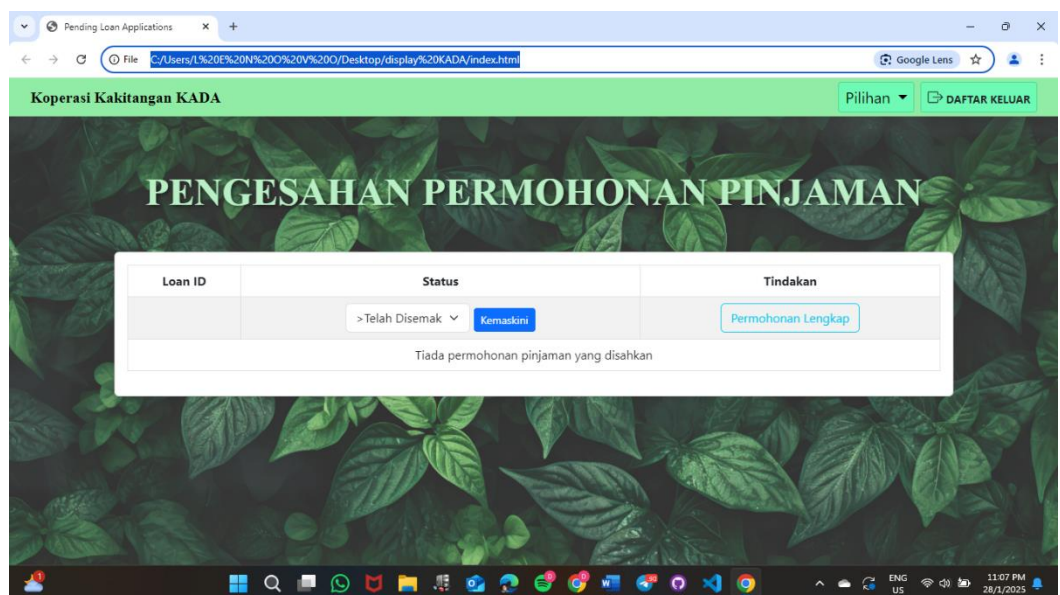
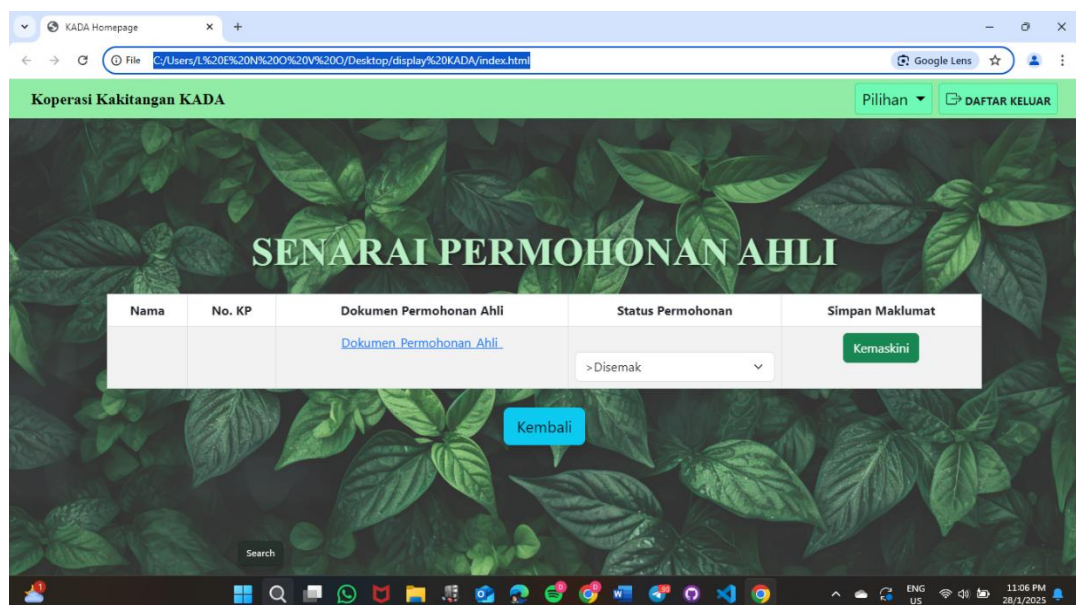
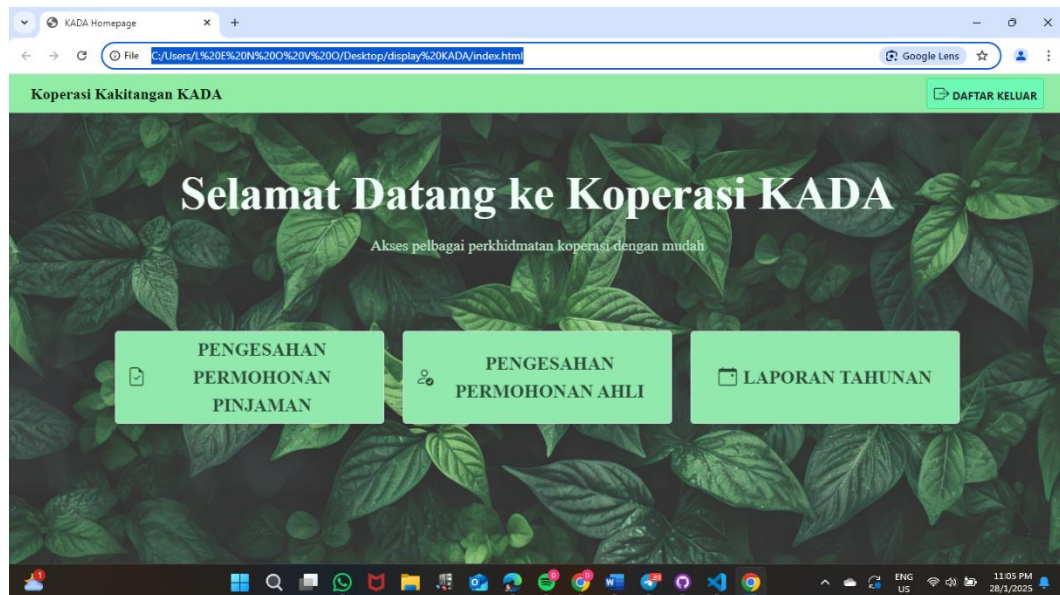
7. Retrieve Loan Details with Member Information

```
SELECT l.loan_id, m.name, l.loan_type, l.loan_amount,
l.repayment_period_months
FROM Loan_Application l
JOIN Member_Application m ON l.member_id = m.applicant_id;
```

8. Retrieve Members with Salary Above a Threshold

```
SELECT name, monthly_salary
FROM Member_Application
WHERE monthly_salary > 7000;
```

3.2 Interface Implementation



SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

In the development of the Membership and Loan Application System for Koperasi KADA, significant achievements have been realized. The module successfully addresses the inefficiencies of manual processes by introducing automation for member registration, loan application, and payment tracking. This has ensured more organized, accurate, and secure data handling, resulting in faster processing times and improved member satisfaction. By centralizing these functions, the system has made it easier to access and manage information, creating a more reliable platform for both the members and the organization.

To further enhance the system in the future, several improvements can be made:

1. Developing a mobile app version of the system could improve accessibility for members, allowing them to apply for memberships or loans, track applications, and make payments directly from their phones.
2. Implementing data analytics tools can provide insights into member needs and loan trends, helping the organization make better decisions and improve services.
3. Simplifying the design of the system's interface can make it more user-friendly, especially for members who are not tech-savvy.

4.2 Reflection

Throughout the process of developing the Membership and Loan Application System, I gained valuable insights and skills that have broadened my understanding of system development.

First, I learned the importance of time management and working under a deadline. Developing a system within a given timeframe was challenging, but it taught me how to prioritize tasks, manage resources effectively, and stay focused to ensure timely delivery. This experience has given me a deeper appreciation for the effort and dedication required to complete a system.

Next, I realized the critical role of teamwork in achieving success. Collaborating with my team members made it clear that effective communication, mutual support,

and sharing responsibilities are essential to handle complex tasks and achieve a common goal. The synergy of the team played a significant role in overcoming challenges and ensuring the project's success.

Additionally, I understood how crucial it is to meet client requirements. Understanding and addressing the client's needs are fundamental to designing a user-friendly system that meets their expectations. This taught me the importance of active listening, thorough requirement gathering, and considering the end user's perspective to deliver a system that is practical and efficient.

Finally, I recognized the importance of technical skills, particularly coding, in system development. Writing and debugging code to bring the system to life was a demanding yet rewarding task. It emphasized how essential technical expertise is in turning ideas into functional and reliable software.

Overall, this experience has been both a learning journey and a confidence booster. It not only enhanced my technical skills but also strengthened my soft skills, such as communication, teamwork, and adaptability. These lessons will undoubtedly help me in future projects, whether in academic settings or professional environments.